TopSecret - Entity Relationship Diagram (Homework 6)
SQLite database: database/missions.db PK = primary key, FK = foreign key

Not in the database:
user accounts and passwords are kept
ciphered in users.cip (one line per user).
audit_log.username, messages.sender and
user_access.username refer to those users.
A user reads the messages of the agent
whose name matches their username (any case).

user_access	
PK username	TEXT
level	TEXT NOT NULL (ADMIN or STANDARD)

audit_log	
PK id	INTEGER AUTOINCREMENT
timestamp	TEXT NOT NULL
username	TEXT NOT NULL
action	TEXT NOT NULL (READ, CREATE, UPDATE, DELETE)
item	TEXT NOT NULL

missions	
PK id	INTEGER AUTOINCREMENT
title	TEXT NOT NULL
brief	TEXT NOT NULL
date	TEXT NOT NULL
UNIQUE (title, date)	

mission_briefs	
PK mission_id	INTEGER NOT NULL (checked in code)
PK, FK brief_id	INTEGER UNIQUE, CASCADE

mission_agents	
PK, FK mission_id	INTEGER, CASCADE
PK, FK agent_id	INTEGER, RESTRICT

briefs	
PK id	INTEGER
title	TEXT NOT NULL
date	TEXT NOT NULL
text	TEXT NOT NULL

mission_facilities	
PK, FK mission_id	INTEGER, CASCADE
FK facility_id	INTEGER NOT NULL, RESTRICT

facilities	
PK id	INTEGER
name	TEXT NOT NULL
abbreviation	TEXT NOT NULL

agents	
PK id	INTEGER
name	TEXT NOT NULL
date_of_birth	TEXT
date_of_death	TEXT
notes	TEXT
FK facility_id	INTEGER NOT NULL, RESTRICT

messages	
PK id	INTEGER AUTOINCREMENT
FK agent_id	INTEGER NOT NULL, CASCADE
sender	TEXT NOT NULL (username)
text	TEXT NOT NULL
sent_at	TEXT NOT NULL

1 : many

1 : 0..many briefs

1 : 0..1
one facility runs a mission

1 : 0..1
(a brief is in at most one mission)

1 : many
(joint operations: any facility)

1 : many

1 : many
every agent has one facility

1 : many